Linear Algebra 2026 Homework 4

Due: October 23, 2026

Problems should be solved in English, and calculation steps are required. For all T/F questions, you don't need to justify your answer.

All notations/theorems can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.

Problem 1 [6 pts]. Label the following statements as true or false. Assume that V and W are finite-dimensional vector spaces with ordered bases β and γ, respectively, and $\mathsf{T}, \mathsf{U}\colon \mathsf{V} \to \mathsf{W}$ are linear transformations.

(a) For any scalar a, $a\mathsf{T} + \mathsf{U}$ is a linear transformation from V to W.

(b) $[\mathsf{T}]_\beta^\gamma = [\mathsf{U}]_\beta^\gamma$ implies that $\mathsf{T} = \mathsf{U}$.

(c) If $m = \dim(\mathsf{V})$ and $n = \dim(\mathsf{W})$, then $[\mathsf{T}]_\beta^\gamma$ is an $m \times n$ matrix.

(d) $[\mathsf{T} + \mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma + [\mathsf{U}]_\beta^\gamma$.

(e) $\mathcal{L}(\mathsf{V}, \mathsf{W})$ is a vector space.

(f) $\mathcal{L}(\mathsf{V}, \mathsf{W}) = \mathcal{L}(\mathsf{W}, \mathsf{V})$.

Problem 2 [12 pts]. Let β and γ be the standard ordered bases for R^n and R^m, respectively. For each linear transformation $\mathsf{T}\colon \mathsf{R}^n \to \mathsf{R}^m$, compute $[\mathsf{T}]_\beta^\gamma$.

(a) $\mathsf{T}\colon \mathsf{R}^2 \to \mathsf{R}^3$ defined by $\mathsf{T}(a_1, a_2) = (2a_1 - a_2,\ 3a_1 + 4a_2,\ a_1)$.

(b) $\mathsf{T}\colon \mathsf{R}^3 \to \mathsf{R}^2$ defined by $\mathsf{T}(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3,\ a_1 + a_3)$.

(c) $\mathsf{T}\colon \mathsf{R}^3 \to \mathsf{R}$ defined by $\mathsf{T}(a_1, a_2, a_3) = 2a_1 + a_2 - 3a_3$.

Problem 3 [10 pts]. Define

$$\mathsf{T}\colon \mathsf{M}_{2\times 2}(\mathsf{R}) \to \mathsf{P}_2(\mathsf{R}) \quad \text{by} \quad \mathsf{T}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a+b) + (2d)x + bx^2.$$

Let

$$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}, \qquad \gamma = \{1, x, x^2\}.$$

Compute $[\mathsf{T}]_\beta^\gamma$.

Problem 4 [10 pts]. Let

$$\alpha = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}, \qquad \beta = \{1, x, x^2\}, \qquad \gamma = \{1\}.$$

Define

$$\mathsf{T}\colon \mathsf{P}_2(\mathsf{R}) \to \mathsf{M}_{2\times 2}(\mathsf{R}) \quad \text{by} \quad \mathsf{T}(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix},$$

where $'$ denotes differentiation. Compute $[\mathsf{T}]_\beta^\alpha$.

Problem 5 [10 pts]. Let V be an n-dimensional vector space with an ordered basis β. Define $\mathsf{T}\colon \mathsf{V} \to F^n$ by $\mathsf{T}(x) = [x]_\beta$. Prove that T is linear.

Problem 6 [10 pts]. Let V be an n-dimensional vector space, and let $\mathsf{T}\colon \mathsf{V} \to \mathsf{V}$ be a linear transformation. Suppose that W is a T-invariant subspace of V (see the exercises of Section 2.1) having dimension k. Show that there is a basis β for V such that $[\mathsf{T}]_\beta$ has the form

$$\begin{pmatrix} A & B \\ O & C \end{pmatrix},$$

where A is a $k \times k$ matrix and O is the $(n-k) \times k$ zero matrix.

Problem 7 [10 pts]. Let V and W be vector spaces, and let T and U be nonzero linear transformations from V into W. If $\mathsf{R}(\mathsf{T}) \cap \mathsf{R}(\mathsf{U}) = \{0\}$, prove that $\{\mathsf{T}, \mathsf{U}\}$ is a linearly independent subset of $\mathcal{L}(\mathsf{V}, \mathsf{W})$.

Problem 8 [15 pts]. Let V and W be vector spaces, and let S be a subset of V. Define

$$S^0 = \{\mathsf{T} \in \mathcal{L}(\mathsf{V}, \mathsf{W}) : \mathsf{T}(x) = 0 \text{ for all } x \in S\}.$$

Prove the following statements.

(a) S^0 is a subspace of $\mathcal{L}(\mathsf{V}, \mathsf{W})$.

(b) If S_1 and S_2 are subsets of V and $S_1 \subseteq S_2$, then $S_2^0 \subseteq S_1^0$.

(c) If V_1 and V_2 are subspaces of V, then $(\mathsf{V}_1 + \mathsf{V}_2)^0 = \mathsf{V}_1^0 \cap \mathsf{V}_2^0$.

Problem 9 [10 pts]. Let V and W be vector spaces such that $\dim(\mathsf{V}) = \dim(\mathsf{W})$, and let $\mathsf{T}\colon \mathsf{V} \to \mathsf{W}$ be linear. Show that there exist ordered bases β and γ for V and W, respectively, such that $[\mathsf{T}]_\beta^\gamma$ is a diagonal matrix.

Problem 10 [10 pts]. Label the following statements as true or false. In each part, V, W, and Z denote vector spaces with ordered (finite) bases α, β, and γ, respectively; $\mathsf{T}\colon \mathsf{V} \to \mathsf{W}$ and $\mathsf{U}\colon \mathsf{W} \to \mathsf{Z}$ denote linear transformations; and A and B denote matrices.

(a) $[\mathsf{UT}]_\alpha^\gamma = [\mathsf{T}]_\alpha^\beta [\mathsf{U}]_\beta^\gamma$.

(b) $[\mathsf{T}(v)]_\beta = [\mathsf{T}]_\alpha^\beta [v]_\alpha$ for all $v \in \mathsf{V}$.

(c) $[\mathsf{U}(w)]_\beta = [\mathsf{U}]_\alpha^\beta [w]_\beta$ for all $w \in \mathsf{W}$.

(d) $[\mathsf{I}_\mathsf{V}]_\alpha = \mathsf{I}$.

(e) $[\mathsf{T}^2]_\alpha^\beta = ([\mathsf{T}]_\alpha^\beta)^2$.

(f) $A^2 = \mathsf{I}$ implies that $A = \mathsf{I}$ or $A = -\mathsf{I}$.

(g) $\mathsf{T} = \mathsf{L}_A$ for some matrix A.

(h) $A^2 = O$ implies that $A = O$, where O denotes the zero matrix.

(i) $\mathsf{L}_{A+B} = \mathsf{L}_A + \mathsf{L}_B$.

(j) If A is square and $A_{ij} = \delta_{ij}$ for all i and j, then $A = \mathsf{I}$.

Problem 11 [16 pts].

(a) Let

$$A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & -3 \\ 4 & 1 & 2 \end{pmatrix},$$

$$C = \begin{pmatrix} 1 & 1 & 4 \\ -1 & -2 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}.$$

Compute $A(2B + 3C)$, $(AB)D$, and $A(BD)$.

(b) Let

$$A = \begin{pmatrix} 2 & 5 \\ -3 & 1 \\ 4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & -2 & 0 \\ 1 & -1 & 4 \\ 5 & 5 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 4 & 0 & 3 \end{pmatrix}.$$

Compute A^t, A^tB, BC^t, CB, and CA.

Problem 12 [20 pts]. Let $g(x) = 3 + x$. Let $\mathsf{T}\colon \mathsf{P}_2(\mathsf{R}) \to \mathsf{P}_2(\mathsf{R})$ and $\mathsf{U}\colon \mathsf{P}_2(\mathsf{R}) \to \mathsf{R}^3$ be the linear transformations respectively defined by

$$\mathsf{T}(f(x)) = f'(x)g(x) + 2f(x) \quad \text{and} \quad \mathsf{U}(a + bx + cx^2) = (a + b,\, c,\, a - b).$$

Let β and γ be the standard ordered bases of $\mathsf{P}_2(\mathsf{R})$ and R^3, respectively.

(a) Compute $[\mathsf{U}]_\beta^\gamma$, $[\mathsf{T}]_\beta$, and $[\mathsf{UT}]_\beta^\gamma$ directly. Then use Theorem 2.11 to verify your result.

(b) Let $h(x) = 3 - 2x + x^2$. Compute $[h(x)]_\beta$ and $[\mathsf{U}(h(x))]_\gamma$. Then use $[\mathsf{U}]_\beta^\gamma$ from (a) and Theorem 2.14 to verify your result.

Problem 13 [20 pts]. Let α and β be the ordered bases given in Problem 4. For each of the following parts, use Theorem 2.14 to compute the indicated vector for the linear transformation T defined below.

(a) $[\mathsf{T}(A)]_\alpha$, where

$$A = \begin{pmatrix} 1 & 4 \\ -1 & 6 \end{pmatrix}.$$

Here $\mathsf{T}\colon \mathsf{M}_{2\times 2}(F) \to \mathsf{M}_{2\times 2}(F)$ is defined by $\mathsf{T}(A) = A^t$.

(b) $[\mathsf{T}(f(x))]_\alpha$, where $f(x) = 4 - 6x + 3x^2$. Here $\mathsf{T}\colon \mathsf{P}_2(\mathsf{R}) \to \mathsf{M}_{2\times 2}(\mathsf{R})$ is defined by

$$\mathsf{T}(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix}.$$

Problem 14 [10 pts]. Find linear transformations $\mathsf{U}, \mathsf{T}\colon F^2 \to F^2$ such that $\mathsf{UT} = \mathsf{T}_0$ (the zero transformation) but $\mathsf{TU} \neq \mathsf{T}_0$. Use your answer to find matrices A and B such that $AB = O$ but $BA \neq O$.

Problem 15 [15 pts]. Let V, W, and Z be vector spaces, and let $\mathsf{T}\colon \mathsf{V} \to \mathsf{W}$ and $\mathsf{U}\colon \mathsf{W} \to \mathsf{Z}$ be linear.

(a) Prove that if UT is one-to-one, then T is one-to-one. Must U also be one-to-one?

(b) Prove that if UT is onto, then U is onto. Must T also be onto?

(c) Prove that if U and T are one-to-one and onto, then UT is also.

Problem 16 [15 pts]. Let V be a finite-dimensional vector space, and let $\mathsf{T}\colon \mathsf{V} \to \mathsf{V}$ be linear.

(a) If $\operatorname{rank}(\mathsf{T}) = \operatorname{rank}(\mathsf{T}^2)$, prove that $\mathsf{R}(\mathsf{T}) \cap \mathsf{N}(\mathsf{T}) = \{0\}$. Deduce that $\mathsf{V} = \mathsf{R}(\mathsf{T}) \oplus \mathsf{N}(\mathsf{T})$ (see the exercises of Section 1.3).

(b) Prove that $\mathsf{V} = \mathsf{R}(\mathsf{T}^k) \oplus \mathsf{N}(\mathsf{T}^k)$ for some positive integer k.

Problem 17 [9 pts]. Label the following statements as true or false. In each part, V and W are vector spaces with ordered (finite) bases α and β, respectively, $\mathsf{T}\colon \mathsf{V} \to \mathsf{W}$ is linear, and A and B are matrices.

(a) $\left([\mathsf{T}]_\alpha^\beta\right)^{-1} = [\mathsf{T}^{-1}]_\alpha^\beta$.

(b) T is invertible if and only if T is one-to-one and onto.

(c) $\mathsf{T} = \mathsf{L}_A$, where $A = [\mathsf{T}]_\alpha^\beta$.

(d) $\mathsf{M}_{2\times 3}(F)$ is isomorphic to F^5.

(e) $\mathsf{P}_n(F)$ is isomorphic to $\mathsf{P}_m(F)$ if and only if $n = m$.

(f) $AB = \mathsf{I}$ implies that A and B are invertible.

(g) If A is invertible, then $(A^{-1})^{-1} = A$.

(h) A is invertible if and only if L_A is invertible.

(i) A must be square in order to possess an inverse.

Problem 18 [15 pts]. For each of the following linear transformations T, determine whether T is invertible and justify your answer.

(a) $\mathsf{T}\colon \mathsf{R}^2 \to \mathsf{R}^3$ defined by $\mathsf{T}(a_1, a_2) = (a_1 - 2a_2,\ a_2,\ 3a_1 + 4a_2)$.

(c) $\mathsf{T}\colon \mathsf{R}^3 \to \mathsf{R}^3$ defined by $\mathsf{T}(a_1, a_2, a_3) = (3a_1 - 2a_3,\ a_2,\ 3a_1 + 4a_2)$.

(e) $\mathsf{T}\colon \mathsf{M}_{2\times 2}(\mathsf{R}) \to \mathsf{P}_2(\mathsf{R})$ defined by

$$\mathsf{T}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + 2bx + (c + d)x^2.$$

Problem 19 [5 pts]. Let A be invertible. Prove that A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$.

Problem 20 [10 pts]. Let A and B be $n \times n$ matrices such that AB is invertible. Prove that A and B are invertible. Give an example to show that arbitrary matrices A and B need not be invertible if AB is invertible.

Problem 21 [10 pts]. For an arbitrary $n \times n$ matrix $A \in \mathsf{M}_{n\times n}(\mathsf{R})$ with all entries smaller than 1 for their absolute values, can we have

$$\lim_{m\to\infty} A^m \to 0$$

Justify your answer.

Problem 22 [10 pts]. Let U be a subspace of a finite-dimensional vector space V. Let $\mathsf{T}\colon \mathsf{V} \to \mathsf{V}$ be a linear transformation. Prove

$$\dim(\mathsf{U}) - \dim\big(\mathsf{N}(\mathsf{T})\big) \le \dim\big(\mathsf{T}(\mathsf{U})\big) \le \dim(\mathsf{U}).$$

Problem 23 [10 pts]. Prove

$$\operatorname{rank}(\mathsf{L}_A) + \operatorname{rank}(\mathsf{L}_B) - n \le \operatorname{rank}(\mathsf{L}_{AB}) \le \min\big\{\operatorname{rank}(\mathsf{L}_A),\, \operatorname{rank}(\mathsf{L}_B)\big\},$$

where A and B are $n \times n$ matrices. *Hint:* Use the dimension inequality in Problem 22.

Problem 24 [10 pts]. Prove

$$\operatorname{rank}(\mathsf{L}_{A+B}) \le \operatorname{rank}(\mathsf{L}_A) + \operatorname{rank}(\mathsf{L}_B)$$

where A and B are $n \times m$ matrices.